Landscape Health Monitoring at Humber Arboretum

DE-BUG GANG

Research Report - Spring 2026

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Introduction

The Humber Arboretum is a public garden and natural landscape located within the Humber River watershed in North Etobicoke, Toronto. Covering approximately 100 hectares (250 acres), the arboretum serves as a living laboratory for students, an environmental education center, and a conservation area that supports local biodiversity.

The landscape includes a variety of natural and designed environments such as deciduous forests, meadow and shrubland areas, wetlands, ponds, river valley terrain, and botanical gardens. These ecosystems support a wide range of plant and animal species, including birds, amphibians, fish, and pollinating insects. Several educational gardens are also located within the arboretum, such as the Garden of the Rising Moon, the Tranquility Bird Garden, and the Pollinator Garden, which combine ecological restoration with environmental education.

Because of this ecological diversity, the arboretum plays an important role in conservation, biodiversity protection, and environmental education. Managing and restoring these landscapes requires continuous monitoring of environmental conditions, vegetation health, and biodiversity. However, collecting and organizing this data can be challenging for the stakeholders responsible for maintaining the landscape.

Therefore, this research explores how a digital tool could support stakeholders in assessing and monitoring landscape health in a way that is accessible, reliable, and cost-effective.

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Main Research Question

How might we support stakeholders in assessing and monitoring landscape health through an accessible, reliable, easy-to-use, and low-cost digital solution that fits within their real working context?

To answer this question, the research is structured around five sub-questions:

- **Stakeholders and Users**
  Who are the key stakeholders involved in monitoring landscape health, and what are their specific needs, workflows, and challenges?
- **Environmental Context**
  What environmental characteristics and ecological indicators are relevant for assessing landscape health in areas such as the Humber Arboretum?
- **Current Monitoring Methods**
  How is landscape health currently monitored, and what digital or manual tools are used for collecting, storing, and analyzing environmental data?
- **Cost and Feasibility Constraints**
  What financial, technical, and maintenance constraints must be considered when developing an affordable and sustainable landscape monitoring tool?
- **Technological Opportunities**
  What technologies could support affordable spatial mapping and monitoring of landscape health?

These five sub-questions address the key dimensions of the project: users, environmental context, current monitoring practices, feasibility constraints, and technological opportunities.

Stakeholders and Users

Who are the key stakeholders involved in monitoring landscape health, and what are their specific needs, workflows, and challenges?

For this project, several stakeholders are involved in monitoring and managing the Humber Arboretum landscape. These stakeholders include:

- Humber Office of Sustainability
- Humber Arboretum staff
- Toronto Region Conservation Authority
- Landscape students
- Conservation groups
- Citizen scientists
- Public visitors
- Schools

While multiple stakeholders interact with the landscape, the **Humber Office of Sustainability** has been identified as the primary stakeholder for this project because they are responsible for sustainability reporting and environmental management at Humber. In addition, public visitors were selected as the project's secondary stakeholder because there is a strong opportunity to increase community engagement within the Arboretum.

Environmental Context – Toronto Ecosystem and Restoration

What environmental characteristics and ecological indicators are relevant for assessing landscape health in areas such as the Humber Arboretum?

About Humber Arboretum

As previously mentioned, the Humber Arboretum is located within the Greater Toronto Area, a highly urbanized metropolitan region characterized by extensive ravine systems, river valleys, wetlands, forests, and shoreline ecosystems connected to Lake Ontario. Historically, urban development in Toronto has caused significant ecological degradation, including habitat fragmentation, wetland loss, river channelization, and declining biodiversity.

In response, ecological restoration has become an important component of municipal and regional environmental planning. Key organizations involved in these efforts include the **City of Toronto**, the **Toronto and Region Conservation Authority (TRCA)**, and **Waterfront Toronto**. Their initiatives aim to restore ecological processes while also supporting climate adaptation and urban resilience.

Ecological Restoration Principles

Restoration initiatives in Toronto are guided by the principles of **ecological restoration and urban ecology**.

Ecological restoration focuses on assisting ecosystem recovery through actions such as:

- reintroducing native species
- removing invasive species
- restoring natural hydrological systems
- enhancing biodiversity
- improving long-term ecosystem resilience

Urban ecology highlights that restoration within cities must be multifunctional. Natural systems must coexist with infrastructure, recreation, and public safety requirements. These approaches are often described as **nature-based solutions**, where ecosystems provide environmental benefits such as flood protection, heat reduction, water purification, and wildlife habitat.

Restoration in Practice

Toronto applies restoration strategies across multiple spatial scales.

Large-scale projects such as the **Don Mouth Naturalization Project** redesign river systems to restore natural hydrological processes while providing flood protection. Similarly, **Tommy Thompson Park** demonstrates long-term habitat development through wetland creation and adaptive ecological management.

Urban ecological corridors such as the **Don Valley**, **Humber River**, and **High Park** are important restoration areas. Restoration activities in these landscapes include invasive species removal, native planting, erosion control, wetland rehabilitation, and prescribed burns to restore oak savannah ecosystems.

At a regional scale, conservation landscapes such as **Rouge National Urban Park** demonstrate large-scale restoration efforts that include forest regeneration, meadow restoration, and protection of species at risk.

Implications for Landscape Monitoring

The success of ecological restoration initiatives depends on **continuous monitoring and adaptive management**. Monitoring environmental indicators such as vegetation health, biodiversity levels, habitat connectivity, and ecosystem conditions is necessary to evaluate restoration progress and inform future management decisions.

Because these landscapes exist within urban environments, monitoring systems must also be **accessible, practical, and usable by multiple stakeholders**, including sustainability offices, conservation authorities, researchers, and public users.

Conclusion

The Humber Arboretum exists within a complex urban ecosystem that has been affected by past development and environmental changes. In response, organizations in Toronto are working to restore natural habitats through actions such as planting native species, removing invasive plants, and improving ecosystem health.

To understand whether these restoration efforts are successful, landscapes need to be monitored over time. Tracking indicators such as vegetation health, biodiversity, and habitat conditions helps stakeholders evaluate progress and make better management decisions.

This highlights the need for practical and accessible tools that allow stakeholders to collect and organize landscape data efficiently. Such tools can support ongoing monitoring efforts and help ensure that restoration projects contribute to long-term environmental sustainability.

Current Monitoring Methods

How is landscape health currently monitored, and what digital or manual tools are used for collecting, storing, and analyzing environmental data?

Understanding existing monitoring tools and platforms can help identify how environmental data is currently collected and managed. Analyzing these solutions provides insight into how organizations monitor landscapes and helps identify features that may be useful for developing a new monitoring system.

To explore current solutions, this research used an **available product analysis method**. This method focuses on examining existing products and tools that address similar problems in order to understand how they work, what problems they solve, and what limitations they may have (Hbo-I, n.d.). By studying these systems, it is possible to identify design patterns and opportunities that may inform the development of a new solution.

Several platforms and tools currently support environmental monitoring, field data collection, or ecological analysis.

Fulcrum

Fulcrum is a field data collection and workflow management platform designed for environmental and field teams. It allows users to collect structured data directly on-site, including geospatial information, photographs, and environmental observations. The platform also supports offline data collection, which is useful when working in remote areas with limited internet connectivity.

Fulcrum is particularly useful for organizations that need to standardize fieldwork processes and integrate collected data with Geographic Information Systems (GIS) and reporting systems.

HealthyPlan.City

HealthyPlan.City, developed by the University of Toronto, is a platform that focuses on analyzing urban environments through health and equity indicators. The platform allows users to explore city data related to tree canopy cover, greenspace availability, air pollution, and other built-environment factors.

Unlike field data collection platforms, HealthyPlan.City focuses more on **analyzing existing urban data** rather than collecting new environmental observations.

Tree Equity Score Analyzer

The Tree Equity Score Analyzer is a mapping and analysis tool that focuses specifically on tree canopy coverage and environmental equity across cities. It helps visualize areas where tree cover is lacking and highlights how access to urban greenery varies between communities.

Compared to broader environmental monitoring platforms, this tool focuses on a **single environmental indicator**, making it useful for targeted environmental planning and policy decisions.

iNaturalist

iNaturalist is a community-based biodiversity observation platform that allows users to record and share observations of plants and animals. The platform supports citizen science by enabling users to upload photos, identify species, and contribute biodiversity data to a global database.

Unlike professional fieldwork tools such as Fulcrum, iNaturalist focuses more on **public participation and biodiversity observations**, making it valuable for engaging communities in environmental monitoring.

Conclusion

These platforms demonstrate different approaches to environmental monitoring. Some tools focus on professional field data collection and structured workflows, while others emphasize data analysis or community participation.

This comparison highlights the opportunity to design a system that combines structured environmental data collection with accessible tools for visualization and engagement. Such an approach could help stakeholders monitor landscape health more effectively while also supporting community involvement.

Cost and Feasibility Constraints

What financial, technical, and maintenance constraints must be considered when developing an affordable and sustainable landscape monitoring tool?

When developing a landscape monitoring solution for environments such as the Humber Arboretum, several practical constraints must be considered. These include financial limitations, technical requirements, and long-term maintenance needs. Understanding these constraints is important to ensure that any proposed system remains realistic, sustainable, and usable by stakeholders.

Financial Constraints

Environmental monitoring systems can require specialized equipment such as sensors, communication devices, and data storage infrastructure. While many environmental sensors and microcontrollers have become more affordable in recent years, deploying large monitoring networks across a landscape can still involve significant costs.

Stakeholders such as the Office of Sustainability or educational institutions often operate with limited budgets. Therefore, any monitoring solution should prioritize **low-cost technologies, open-source tools, and scalable systems** that can be expanded gradually rather than requiring large initial investments.

Technical Constraints

Another important consideration is the technical complexity of monitoring systems. Advanced monitoring technologies such as sensor networks or IoT-based systems may require specialized technical knowledge for installation, configuration, and troubleshooting.

Stakeholders responsible for managing the system may not always have extensive technical expertise. As a result, monitoring tools should aim to be **simple to use, easy to maintain, and compatible with existing workflows**.

Maintenance and Infrastructure

Environmental monitoring systems deployed in outdoor environments must also account for maintenance and infrastructure challenges. Sensor equipment may be exposed to weather conditions, wildlife interactions, or physical damage. Additionally, systems may require reliable power sources and communication networks to function effectively.

Regular maintenance, battery replacement, and system updates may therefore be necessary to ensure accurate data collection. These requirements should be considered when designing a monitoring solution to avoid creating systems that are difficult to sustain over time.

Conclusion

Considering financial, technical, and maintenance constraints is essential when developing landscape monitoring solutions. Systems that are too complex or expensive may be difficult for stakeholders to adopt and maintain. Therefore, monitoring tools should aim to balance technological capabilities with practical feasibility, ensuring that they remain accessible, affordable, and sustainable for long-term use.

Technological Opportunities

What technologies could support affordable spatial mapping and monitoring of landscape health?

Recent advances in digital technologies provide new opportunities for monitoring environmental conditions in natural landscapes such as the Humber Arboretum. Technologies such as environmental sensors, wireless communication systems, and spatial data platforms can support the collection and visualization of ecological data in a more efficient and automated way.

IoT-Based Environmental Monitoring

One possible approach for monitoring landscape health is the use of **Internet of Things (IoT) systems**. These systems consist of distributed sensor nodes placed throughout a landscape to collect environmental data. The data is then transmitted to a central platform where it can be analyzed and visualized.

Environmental monitoring systems often follow a structure such as:

Sensor Node → Microcontroller → Communication Module → Gateway → Cloud Server → Database → Application

Through this architecture, environmental data such as soil conditions, climate data, water quality, and wildlife activity can be monitored in near real time. The collected data can then be displayed through dashboards or geographic information systems (GIS) for analysis.

Sensor Technologies

Various sensors are used in environmental monitoring to measure indicators related to ecosystem health.

Examples of commonly used sensors include:

Climate monitoring

- Temperature and humidity sensors (BME280, BME680)

Air quality monitoring

- Gas sensors (MQ-135)
- CO_2 sensors (SCD30)

Soil monitoring

- Soil moisture sensors
- Soil temperature sensors (DS18B20)
- Soil pH sensors

Light monitoring

- Light intensity sensors (BH1750)

Water monitoring

- Turbidity sensors
- Water level sensors
- Total dissolved solids (TDS) sensors

These sensors allow environmental conditions to be monitored continuously and can provide useful data about ecosystem health.

Wildlife Monitoring Technologies

Monitoring wildlife activity is also important for understanding biodiversity within an ecosystem. Technologies used for wildlife monitoring include:

- trail cameras
- motion sensors (PIR sensors)
- acoustic microphones for detecting animal sounds
- camera modules such as ESP32-CAM

These technologies can help detect wildlife presence, observe animal behavior patterns, and track changes in biodiversity over time.

Communication Technologies

Environmental sensor networks require communication systems to transmit collected data to a central platform. Several communication technologies are commonly used for this purpose, including:

- WiFi networks
- Cellular LTE communication
- Zigbee mesh networks
- LoRa (Low Power Wide Area Network)

LoRa technology is often considered suitable for environmental monitoring because it allows long-range communication while consuming relatively low power. This makes it useful for monitoring large natural landscapes where infrastructure and electricity may be limited.

Power Systems

Sensor nodes placed outdoors often require autonomous power systems. A common setup includes:

Solar Panel → Charge Controller → Battery → Sensor Node

Solar-powered systems allow monitoring equipment to operate for extended periods with minimal maintenance, which makes them practical for environmental monitoring projects.

Conclusion

The availability of low-cost sensors, microcontrollers, and communication technologies creates opportunities for developing affordable environmental monitoring systems. These technologies could support the collection of ecological data such as climate conditions, soil health, water quality, and wildlife activity.

However, implementing large-scale sensor networks may require significant infrastructure, maintenance, and technical expertise. For this project, these technologies should therefore be considered as potential tools that could support landscape monitoring, depending on stakeholder needs and the practical scope of the solution.

Key Research Insights

The research conducted on the Humber Arboretum, its stakeholders, and current monitoring practices highlights several important insights that guide the direction of this project.

• **Urban ecosystems require continuous monitoring**

Urban natural areas such as the Humber Arboretum are affected by development, climate change, and environmental pressures. Continuous monitoring of vegetation health, biodiversity, and environmental conditions is necessary to evaluate restoration efforts and maintain ecosystem health.

• **Multiple stakeholders are involved in landscape monitoring**

Landscape monitoring involves stakeholders such as the Humber Office of Sustainability, arboretum staff, conservation authorities, students, and public visitors. Monitoring tools must therefore be accessible and usable by diverse user groups.

• **Current monitoring methods are often fragmented**

Existing tools focus on different aspects of environmental monitoring, such as professional field data collection, urban data analysis, or citizen science observations. However, these systems are often disconnected and do not provide a unified monitoring platform.

• **Practical constraints must be considered**

Monitoring systems must remain affordable, technically manageable, and maintainable over time to ensure long-term adoption by stakeholders.

• **New technologies create opportunities for improved monitoring**

Recent developments in sensors, wireless communication systems, and digital mapping platforms provide opportunities for more efficient environmental monitoring.

Problem Statement Definition

Humber Office of Sustainability

The Humber Office of Sustainability needs a way to systematically monitor, collect, record, and analyze data on landscape elements across the Humber Arboretum and surrounding campus properties, because they currently lack a centralized and reliable system to track ecological metrics and landscape features such as green space coverage.

This is important because Humber College has committed to achieving Net-Zero emissions by 2029, and without accurate, ongoing, and reportable landscape data, the Office cannot effectively measure progress, identify gaps, make evidence-based land management decisions, or produce credible sustainability reports for institutional leadership, funders, and the broader community.

Public Visitors

Public visitors to the Humber Arboretum need a way to access meaningful, easy-to-understand information about the ecological features and sustainability efforts across the Arboretum grounds, because they currently have limited visibility into the landscape elements around them, such as tree species, biodiversity, and the College's broader environmental initiatives and have no accessible way to engage with or contribute to the sustainability story of the space they are walking through.

This is important because without an engaging and informative visitor experience, opportunities to build public environmental awareness, foster community stewardship, and inspire broader support for sustainability are lost, reducing the Arboretum's impact beyond its ecological function alone. Additionally, the support for hobbyists is limited and the value of the application for less technically oriented people is lost.

Bibliography

Fulcrum. (2026, February 26). *Environmental | Field data collection and process management*. https://www.fulcrumapp.com/industries/environmental/

HealthyPlan. (n.d.). https://healthyplan.city/en

Arboretum, H. (n.d.). *About the Humber arboretum*. https://humber.ca/arboretum/about.html

Botanischer Garten::Institut für Botanik::Department für Ökosystemmanagement, Klima und Biodiversität::BOKU. (n.d.). https://boku.ac.at/oekb/bota/einrichtungen/botanischer-garten

Botanischer Garten Bodenkultur Wien. (n.d.). https://botanischergarten.univie.ac.at/netzwerk/organisationen/arge-oesterreichischer-botanischer-gaerten/botanischer-garten-bodenkultur-wien/

TESA Toronto. (n.d.). https://www.treeequityscore.org/analyzer/toronto/map

INaturalist. (n.d.). iNaturalist. https://www.inaturalist.org/?utm_source=google_cpc&utm_medium=ad_grant&utm_campaign=cbc_ggrant_Brand_Eng_Max_Clicks&gad_source=1&gad_campaignid=23370780629&gbraid=0AAAABAxkD6vEEYZbLz0fS_Ju2r0SpL0fm&gclid=Cj0KCQjw37nNBhDkARIsAEBGI8Oyzt5FsGxZGWana6xbI3Iptxc6zUP-fQXVoOTgTzovGPEOdvISDRkaAm2IEALw_wcB

Pfister, Z., Vierhauser, M., Medvedova, A., Schroeder, M., Rampp, M., Kronenberg, A., ... & Simon, A. (2025). FORTE: an open-source system for cost-effective and scalable environmental monitoring. arXiv preprint arXiv:2502.00049https://arxiv.org/abs/2502.00049

Li, Z., Li, X., & Shang, J. (2025). Forest Fire Monitoring and Energy Optimization Based on LoRa-Mesh Wireless Communication Technology. Electronics, 14(21), 4135. https://doi.org/10.3390/electronics14214135

Sharma, A., Kapoor, D.S., Nayyar, A., Qureshi, B., Singh, K.J. et al. (2022). Exploration of IoT Nodes Communication Using LoRaWAN in Forest Environment. Computers, Materials & Continua, 71(3), 6239–6256. https://doi.org/10.32604/cmc.2022.024639

Hart, J. K., & Martinez, K. (2006). Environmental sensor networks: A revolution in the Earth system science? Earth-Science Reviews, 78(3–4), 177–191. https://doi.org/10.1016/j.earscirev.2006.05.001